Lecture 23: Quadrature Examples; Gaussian Quadrature

Math 6610

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Today's Goals

- B-splines as a basis: de Boor points, local control
- Interpolation in B-spline form: banded system, conditioning
- Compound Simpson: reuse of A-sums and B-sums
- Gaussian quadrature: $2n$ parameters $\Rightarrow$ aim for degree $2n-1$
- Nodes: zeros of the nth orthogonal (Legendre) polynomial
- Weights: exactness on $\prod_{n-1}$

Recall: Spline Space and B-Splines

$\Delta : a = x_0 < x_1 < \cdots < x_n = b$, order $k \geqslant 2$. $\mathcal{S}_{k,\Delta}$ is the set of $S \in C^{k-2}[a,b]$ that lie in $\prod_{k-1}$ on each $[x_{j-1}, x_j]$; $\dim \mathcal{S}_{k,\Delta} = k + n - 1$.

Clamped knots: $\tau_1 = \cdots = \tau_k = x_0$, $\tau_{k+j} = x_j$ for $j = 1, \ldots, n-1$, $\tau_{k+n} = \cdots = \tau_{2k+n-1} = x_n$; so $l = 2k + n - 1$. The B-splines N_{ik} come from the Cox–de Boor recursion on $\tau_1 \leqslant \cdots \leqslant \tau_l$.

Proposition (*proofs omitted, de Boor, Ch. IX*). For clamped knots and $1 \leqslant i \leqslant l - k$:

(a) $N_{ik}(x) = 0$ for $x \notin [\tau_i, \tau_{i+k})$;

(b) $N_{ik}(x) \geqslant 0$ for all x;

(c) $\sum_{i=1}^{l-k} N_{ik}(x) = 1$ for all $x \in [a,b]$;

(d) $N_{ik} \in \mathcal{S}_{k,\Delta}$.

Clamped Knots Give a Basis

Theorem (*proof omitted; de Boor, Ch. IX*). For clamped knots ($l = 2k + n - 1$), the B-splines N_{ik}, $i = 1, \ldots, k + n - 1$, form a basis of $\mathcal{S}_{k,\Delta}$.

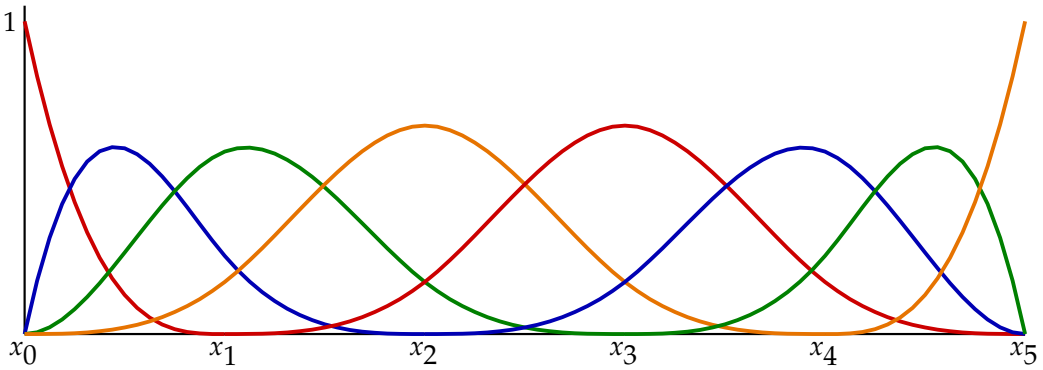
Remark (*Why it is plausible*). Each $N_{ik} \in \mathcal{S}_{k,\Delta}$ by (d), and there are $k + n - 1 = \dim \mathcal{S}_{k,\Delta}$ of them. What remains is linear independence.

So every $S \in \mathcal{S}_{k,\Delta}$ is, in exactly one way,

$$S(x) = \sum_{i=1}^{k+n-1} \gamma_i N_{ik}(x).$$

Clamped Cubic B-Splines

$k = 4$, $x_0, \ldots, x_5 = 0, 1, \ldots, 5$: knots $\tau = (0, 0, 0, 0, 1, 2, 3, 4, 5, 5, 5, 5)$, $l = 12$, 8 B-splines.



- End B-splines start at height 1: $N_{14}(x_0) = N_{84}(x_5) = 1$.
- $N_{i4} = 0$ outside $[\tau_i, \tau_{i+4})$ (a); at each x at most 4 are nonzero, and they sum to 1 (c).

De Boor Points

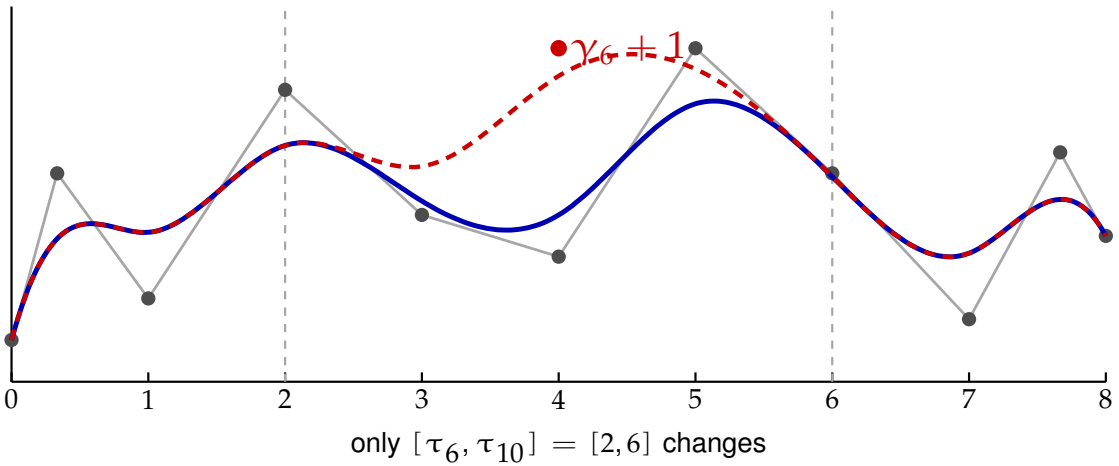
Definition (*de Boor points*). For $S = \sum_{i=1}^{k+n-1} \gamma_i N_{ik} \in \mathcal{S}_{k,\Delta}$, the coefficients $\gamma_1, \ldots, \gamma_{k+n-1}$ are the *de Boor points* of S.

Remark (*What (a)–(c) give*). At each $x \in [a, b]$, $S(x)$ is a weighted average of the γ_i (weights $N_{ik}(x) \geqslant 0$, sum 1). Hence:

- no overshoot: $\min_i \gamma_i \leqslant S(x) \leqslant \max_i \gamma_i$;
- stable: if every γ_i changes by at most δ, then $S(x)$ changes by at most $\delta \sum_i N_{ik}(x) = \delta$;
- local control: changing only γ_i changes S only on $[\tau_i, \tau_{i+k}]$ (by (a)).

De Boor Points: Local Control

Cubic, clamped knots on $0, 1, \ldots, 8$; 11 de Boor points (dots), each drawn at $\xi_i = (\tau_{i+1} + \tau_{i+2} + \tau_{i+3})/3$:



- Blue: S. It follows the polygon of de Boor points, smoothed.
- Red dashed: γ_6 raised by 1. S moves only on $[\tau_6, \tau_{10}] = [2, 6]$; outside, nothing changes.
- A B-spline curve does not pass through its de Boor points: to hit data we must solve for them (next frames).

Interpolation: The Not-a-Knot Cubic

$k = 4$, $n \geqslant 3$. Drop x_1 and x_{n-1} as knots (Lecture 18):

$$\underbrace{x_0}_{\tau_1=\cdots=\tau_4} < x_1 < \underbrace{x_2}_{\tau_5} < \cdots < \underbrace{x_{n-2}}_{\tau_{n+1}} < x_{n-1} < \underbrace{x_n}_{\tau_{n+2}=\cdots=\tau_{n+5}}$$

$l = n + 5$, so $n + 1$ B-splines: $S(x) = \sum_{i=1}^{n+1} \gamma_i N_{i4}(x)$.

Conditions $S(x_{r-1}) = y_{r-1}$, $r = 1, \ldots, n+1$:

$$M\gamma = y, \qquad M = [m_{ri}]_{r,i=1}^{n+1}, \quad m_{ri} = N_{i4}(x_{r-1}).$$

Row r = the B-splines evaluated at one point: at most 4 nonzeros, in consecutive columns (property (a)). Each row sums to 1 by (c).

B-Spline Interpolation in Matlab

Build M row by row, then solve:

```
function [tau, g] = bsinterp(x, y)
% not-a-knot cubic spline through (x_j, y_j), in B-spline form
x = x(:);  y = y(:);  n = length(x) - 1;  k = 4;
tau = [x(1)*ones(1,4), x(3:n-1)', x(n+1)*ones(1,4)];  % l=n+5
l = numel(tau);  M = sparse(n+1, n+1);
for r = 1:n+1
  m = min(find(tau <= x(r), 1, 'last'), l-k);
  M(r, m-k+1:m) = bsplvb(tau, k, x(r), m);
end
g = M \ y;                                    % banded solve
```

- `bsplvb(tau, k, x, m)` (triangular scheme of the B-spline lecture) returns the k nonzero values $N_{m-k+1,k}(x), \ldots, N_{m,k}(x)$ for $\tau_m \leqslant x < \tau_{m+1}$.
- `x(1)` $= x_0$, so `x(3:n-1)` $= x_2, \ldots, x_{n-2}$.
- `min(..., l-k)`: at $x_n = b$, use the last nonempty interval.

B-Spline Interpolation in Matlab

Evaluate $S(x) = \sum_i \gamma_i N_{i4}(x)$: 4 terms per point.

```
function s = bsval(tau, g, xx)
k = 4;  l = numel(tau);  s = zeros(size(xx));
for q = 1:numel(xx)
  m = min(find(tau <= xx(q), 1, 'last'), l-k);
  s(q) = g(m-k+1:m)' * bsplvb(tau, k, xx(q), m);
end
```

Compare with Matlab's `spline` (also not-a-knot):

```
x = linspace(0, 4, 9);  y = sin(x);
[tau, g] = bsinterp(x, y);
xx = linspace(0, 4, 101);
max(abs(bsval(tau, g, xx) - spline(x, y, xx)))   % rounding
    level
```

Same interpolant (it is unique), two representations: piecewise coefficients vs de Boor points.

Solving with M

Cost. At most 4 nonzeros per row, consecutive: M is banded. Storage $O(n)$; banded LU $O(k^2 n)$ flops. (`M \ y` detects the band.)

Remark (*Facts, proofs omitted*).

1. *Schoenberg–Whitney:* M is nonsingular (invertible) if and only if all diagonal entries $N_{i4}(x_{i-1})$ are nonzero. Here they are.
2. M is *totally nonnegative* (every square submatrix has determinant $\geqslant 0$). For such matrices, Gaussian elimination without pivoting is backward stable (de Boor and Pinkus, 1977).

Conclusion: solvable, no pivoting, $O(n)$ work. Same lesson as the natural spline in Lecture 18.

Payoff: B-Splines vs Truncated Powers

Same interpolant (not-a-knot cubic, $n+1$ equally spaced points on $[0,1]$), two bases: T (truncated powers, Lecture 21), M (B-splines).

```
cond(T, inf),  cond(full(M), inf)
```

n	5	10	20	40	80
$\kappa_\infty(T)$	$2.9 \cdot 10^3$	$4.2 \cdot 10^4$	$4.9 \cdot 10^5$	$6.4 \cdot 10^6$	$9.2 \cdot 10^7$
$\kappa_\infty(M)$	4.6	4.8	4.8	4.8	4.8

- $\kappa_\infty(T)$: $\times 14$ per doubling of n (about n^4). $\kappa_\infty(M)$: below 5 for all n.
- Same space, same spline: the basis decides the computation.

Matlab's Own Tools

task	Matlab
not-a-knot cubic, piecewise form	`pp = spline(x, y)`
evaluate piecewise form	`ppval(pp, xx)`
knots with end multiplicity k	`tau = augknt(knots, k)`
matrix M	`M = spcol(tau, k, x)`
interpolant in B-spline form	`sp = spapi(tau, x, y)`
evaluate B-spline form	`fnval(sp, xx)`

(All but `spline`, `ppval`: Curve Fitting Toolbox.)

- Not-a-knot in B-spline form: `tau = augknt(x([1 3:end-2 end]), 4)`.
- `bsplvb` and `bsinterp` are what these do inside.

Summary: Why B-Splines

	truncated powers	B-splines
nonzero terms at a point	$O(n)$	$\leqslant k$
evaluation at a point	$O(n)$	$\frac{7}{2}k(k-1)$ flops
interpolation matrix	dense below diagonal	banded, width k
κ_∞, $k = 4$, $n = 80$	$9.2 \cdot 10^7$	4.8
meaning of coefficients	scaled jumps of $S^{(k-1)}$	de Boor points

Stable to compute, cheap to evaluate, easy to edit locally.

Compound Simpson: Setup

Definition (*Compound Simpson rule*). Let $s \geqslant 1$ be an integer, $h := \frac{b-a}{2s}$, $x_i := a + ih$ for $i = 0, 1, \ldots, 2s$. Simpson's rule on each of the s panels $[x_{2j-2}, x_{2j}]$ (width $2h$) gives

$$S_{2s+1} := \frac{h}{3} \sum_{j=1}^{s} \left[f(x_{2j-2}) + 4f(x_{2j-1}) + f(x_{2j}) \right].$$

It uses $2s + 1$ points.

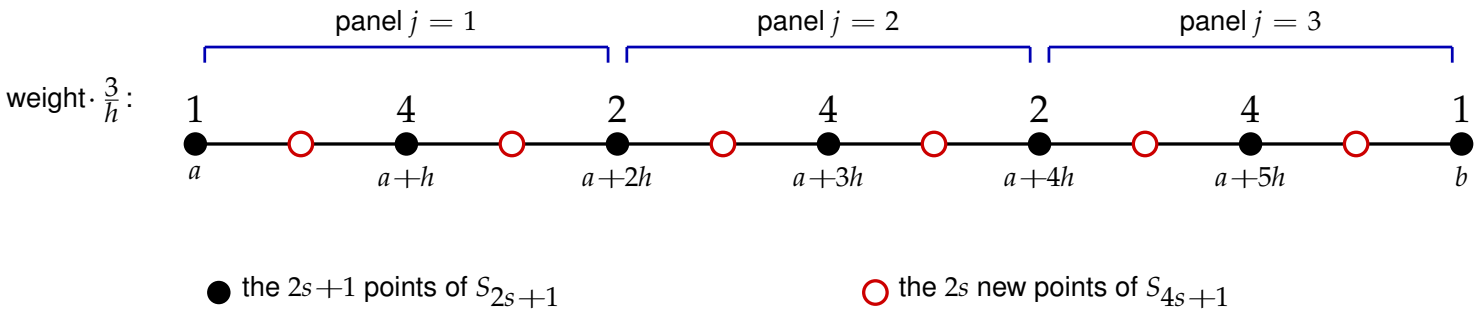


Figure: $s = 3$. Each panel is one Simpson rule. The $2s$ circles are the midpoints of the $2s$ subintervals of width h.

Compound Simpson: Derivation

1. *Split the integral into the s panels.*

$$I = \int_a^b f(x)\,dx = \sum_{j=1}^{s} \int_{x_{2j-2}}^{x_{2j}} f(x)\,dx.$$

2. *Apply Simpson's rule on each panel* (width $2h$, midpoint x_{2j-1}):

$$I \approx \frac{h}{3}\Big[\underbrace{f(x_0)+4f(x_1)+f(x_2)}_{\text{panel 1}} + \underbrace{f(x_2)+4f(x_3)+f(x_4)}_{\text{panel 2}} + \dots$$
$$+ \underbrace{f(x_{2s-2})+4f(x_{2s-1})+f(x_{2s})}_{\text{panel } s}\Big].$$

Compound Simpson: Derivation

3. Collect terms. From step 2, each interior x_{2k} lies in two panels: weight $1 + 1 = 2$.

$$S_{2s+1} = \frac{h}{3}\left[f(a) + 4\sum_{j=1}^{s} f(x_{2j-1}) + 2\sum_{k=1}^{s-1} f(x_{2k}) + f(b)\right],$$

So the weights are $\frac{h}{3}(1, 4, 2, 4, 2, \ldots, 2, 4, 1)$. (For $s = 1$ the second sum is empty.)

Cost: $2s + 1$ evaluations of f (the black dots in the figure of the setup slide).

Compound Simpson: Remainder

Let $R_{2s+1} := S_{2s+1} - I$.

Proposition. If $f \in C^4[a,b]$, there is $\xi \in (a,b)$ with

$$R_{2s+1} = (b-a)\frac{h^4}{180} f^{(4)}(\xi).$$

Proof. *1. One panel* (width $2h$). Simpson's remainder is $\frac{(2h)^5}{2880} f^{(4)}(\xi_j) = \frac{h^5}{90} f^{(4)}(\xi_j)$ for some $\xi_j \in (x_{2j-2}, x_{2j})$.
2. Sum over the s panels.

$$R_{2s+1} = \frac{h^5}{90} \sum_{j=1}^{s} f^{(4)}(\xi_j) = \frac{s\,h^5}{90} \cdot \frac{1}{s} \sum_{j=1}^{s} f^{(4)}(\xi_j).$$

(continued on the next slide)

Compound Simpson: Remainder

Proof (continued). 3. One point ξ. The average $\frac{1}{s}\sum_j f^{(4)}(\xi_j)$ lies between the smallest and the largest of the values $f^{(4)}(\xi_j)$. By the intermediate value theorem for the continuous $f^{(4)}$, it equals $f^{(4)}(\xi)$ for some ξ between the corresponding ξ_j, hence $\xi \in (a, b)$.
4. Constants. Since $s = \frac{b-a}{2h}$,

$$\frac{s\,h^5}{90} = \frac{b-a}{2h} \cdot \frac{h^5}{90} = (b-a)\frac{h^4}{180}.$$

■

Remark. The error is $O(h^4)$ for $f \in C^4$: halving h divides the bound $\frac{b-a}{180}h^4 \max\left|f^{(4)}\right|$ by 16.

Halving h: Error Estimate

Compare S_{2s+1} (step h) with S_{4s+1} (step $\hat{h} = \frac{h}{2} = \frac{b-a}{4s}$). The remainder formula with $\hat{h}$ gives, for some $\hat{\xi} \in (a, b)$,

$$R_{4s+1} = (b-a)\frac{(h/2)^4}{180} f^{(4)}(\hat{\xi}) = \frac{1}{16}(b-a)\frac{h^4}{180} f^{(4)}(\hat{\xi}).$$

Assumption (the only approximation): $f^{(4)}(\xi) \approx f^{(4)}(\hat{\xi})$. Then

$$R_{4s+1} \approx \tfrac{1}{16} R_{2s+1}.$$

Since $S_{4s+1} - S_{2s+1} = R_{4s+1} - R_{2s+1}$,

$$S_{4s+1} - S_{2s+1} \approx -\tfrac{15}{16} R_{2s+1},$$

and therefore

$$|R_{2s+1}| \approx \tfrac{16}{15} |S_{4s+1} - S_{2s+1}|, \qquad |R_{4s+1}| \approx \tfrac{1}{15} |S_{4s+1} - S_{2s+1}|.$$

Remark. Use $I \approx S_{4s+1}$. The quantity $\frac{16}{15} |S_{4s+1} - S_{2s+1}|$ estimates the error of S_{2s+1}, hence is a conservative estimate for the error of S_{4s+1}.

Halving h: Reusing Values

Let $h = \frac{b-a}{2s}$.

Definition (*The sums A_{2s+1} and B_s*).

$$A_{2s+1} := f(a) + 2\sum_{i=1}^{2s-1} f(a+ih) + f(b) \qquad \text{(all points, interior ones doubled)}$$

$$B_s := 2\sum_{j=1}^{s} f\big(a+(2j-1)h\big) \qquad \text{(odd-indexed points)}$$

Proposition. $S_{2s+1} = \dfrac{h}{3}\big(A_{2s+1} + B_s\big)$.

Proof. In $A_{2s+1} + B_s$ the endpoints have weight 1, each even-indexed interior point has weight 2 (from A_{2s+1}), and each odd-indexed point has weight $2+2=4$. This is the weight pattern $(1,4,2,4,\ldots,4,1)$ of S_{2s+1}. ■

Halving h: Reusing Values

Step $\hat{h} = \frac{h}{2}$. The proposition with s replaced by $2s$ gives

$$S_{4s+1} = \frac{\hat{h}}{3}\left(A_{4s+1} + B_{2s}\right) = \frac{b-a}{12s}\left(A_{4s+1} + B_{2s}\right),$$

where

$$A_{4s+1} = f(a) + 2\sum_{i=1}^{4s-1} f(a + i\hat{h}) + f(b),$$

$$B_{2s} = 2\sum_{j=1}^{2s} f\left(a + (2j-1)\hat{h}\right).$$

Proposition. $A_{4s+1} = A_{2s+1} + B_{2s}$.

Proof. Split the sum in A_{4s+1} by the parity of i. For $i = 2k$, $k = 1, \ldots, 2s-1$, we have $a + 2k\hat{h} = a + kh$: these terms form the sum in A_{2s+1}. For $i = 2j-1$, $j = 1, \ldots, 2s$: these terms form B_{2s}. ■

Halving h: One Refinement Step

Remark (*Algorithm*). Given A_{2s+1}, S_{2s+1}, and a tolerance tol:

1. Compute B_{2s}: only $2s$ new evaluations of f (all other values are already in A_{2s+1}).
2. $A_{4s+1} = A_{2s+1} + B_{2s}$.
3. $S_{4s+1} = \frac{b-a}{12s}\left(A_{4s+1} + B_{2s}\right)$.
4. $E = \frac{16}{15}\left|S_{4s+1} - S_{2s+1}\right|$. If $E \leqslant \mathrm{tol}$, accept S_{4s+1}; otherwise replace s by $2s$ and repeat.

Halving h: A Worked Check

Example. $f(x) = x^4$ on $[0,1]$, so $I = \frac{1}{5}$ and $f^{(4)} \equiv 24$.

$$
\begin{aligned}
s = 1,\ h = \tfrac{1}{2}: \quad & A_3 = 0 + 2 \cdot \tfrac{1}{16} + 1 = 1.125, \quad B_1 = 2 \cdot \tfrac{1}{16} = 0.125, \\
& S_3 = \tfrac{1}{6}(1.25) = 0.208333. \\
\hat{h} = \tfrac{1}{4}: \quad & B_2 = 2\left(\tfrac{1}{256} + \tfrac{81}{256}\right) = 0.640625, \\
& A_5 = 1.125 + 0.640625 = 1.765625, \\
& S_5 = \tfrac{1}{12}(A_5 + B_2) = \tfrac{1}{12}(2.40625) = 0.200521.
\end{aligned}
$$

Check: $R_3 = 0.008333 = \frac{1}{120} = (1)\frac{(1/2)^4}{180} \cdot 24$, $R_5 = 0.000521 = \frac{R_3}{16}$, and $\frac{16}{15}|S_5 - S_3| = \frac{16}{15}(0.0078125) = 0.008333$ (exact here because $f^{(4)}$ is constant).

Quadrature Rules and Their Parameters

So far, every formula has the form

$$I = \int_a^b f(x)\,dx \approx \sum_{i=1}^{n} w_i f(x_i). \tag{1}$$

Definition (*Degree of exactness*). Let $\prod_m$ denote the polynomials of degree $\leqslant m$. Rule (1) has *degree of exactness* m if $\sum_i w_i p(x_i) = \int_a^b p(x)\,dx$ for all $p \in \prod_m$, but not for all $p \in \prod_{m+1}$.

Counting. Rule (1) has $2n$ parameters $x_1, \ldots, x_n, w_1, \ldots, w_n$, and $\dim \prod_{2n-1} = 2n$.

Remark (*Heuristic*). Hence we hope to choose the parameters so that the degree of exactness is $2n-1$. Counting alone proves nothing: the exactness equations are nonlinear in the nodes. The choice of nodes in Gaussian quadrature (next frames) makes it work; the proof is in the next lecture.

Gaussian Nodes: Orthogonal Polynomials

Definition (*Inner product and orthogonal polynomial*). On $\prod_\infty$ (polynomials of any degree) let $\langle p, q\rangle := \int_a^b p(x)q(x)\,dx$. The *$n$th orthogonal polynomial* is the monic $p_n(x) = x^n + \cdots \in \prod_n$ with

$$\langle p, p_n\rangle = 0 \qquad \text{for all } p \in \prod_{n-1}.$$

Gauss nodes: $a < x_1 < \cdots < x_n < b$ are the zeros of p_n (they exist: see "The Zeros of p_n" below). For $a = -1$, $b = 1$: $p_n = \gamma_n P_n$ with the Legendre polynomial P_n and γ_n making p_n monic; $p_1 = x,\ p_2 = x^2 - \frac{1}{3},\ p_3 = x^3 - \frac{3}{5}x$.

Orthogonal Polynomials: Construction

Gram–Schmidt on $1, x, x^2, \ldots$*:*

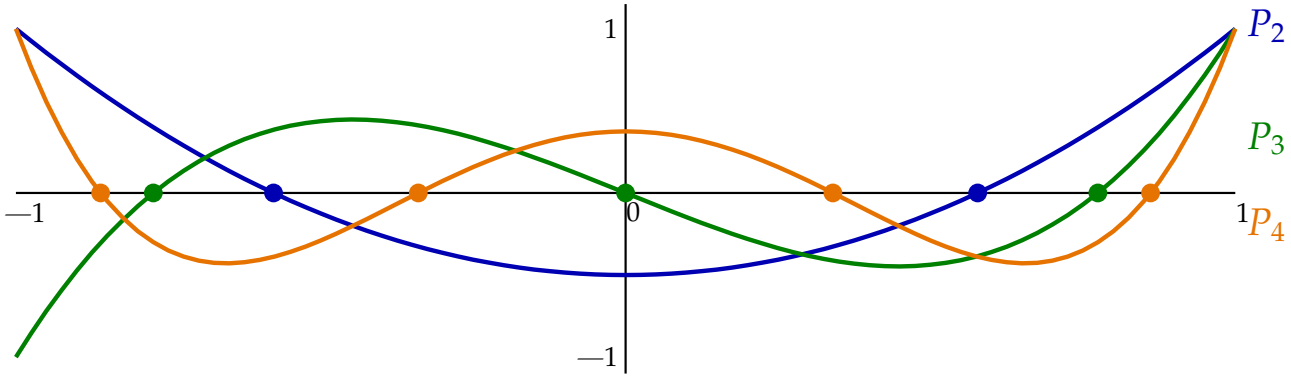
$$p_0 = 1, \qquad p_n = x^n - \sum_{j=0}^{n-1} \frac{\langle x^n, p_j \rangle}{\langle p_j, p_j \rangle} p_j.$$

Inductively $\deg p_j = j$ and $\langle p_j, p_i \rangle = 0$ for $i \neq j$, so for $i < n$: $\langle p_n, p_i \rangle = \langle x^n, p_i \rangle - \langle x^n, p_i \rangle = 0$. As $p_0, \ldots, p_{n-1}$ span $\prod_{n-1}$, p_n is orthogonal to $\prod_{n-1}$.

Remark. Check ($n = 2$): $\langle x^2, p_0 \rangle = \frac{2}{3}$, $\langle p_0, p_0 \rangle = 2$, $\langle x^2, p_1 \rangle = 0$, so $p_2 = x^2 - \frac{1}{3}$.

Orthogonal Polynomials: A Picture

Legendre polynomials P_2, P_3, P_4 on $[-1, 1]$, normalized by $P_n(1) = 1$; dots mark their zeros. (p_n is a constant multiple of P_n, so it has the same zeros.)



- P_n has exactly n zeros, all inside $(-1, 1)$, all distinct.
- P_2: $\pm\frac{1}{\sqrt{3}}$. P_3: $0, \pm\sqrt{\frac{3}{5}}$.

The next frame proves this for every n and every interval $[a, b]$.

The Zeros of p_n Are Usable Nodes

Lemma. For $n \geqslant 1$, p_n has n distinct real zeros, all in (a, b).

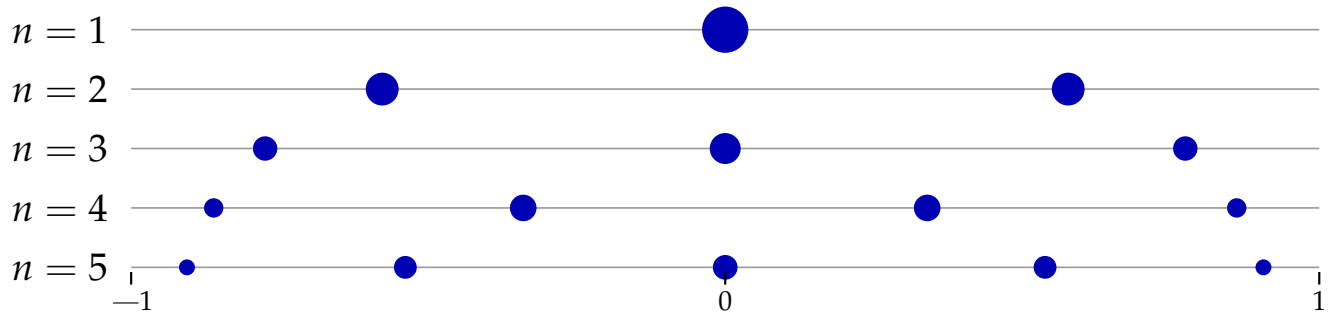
Proof. 1. Count sign changes. Let $x_1, \ldots, x_m$ be the points of (a, b) where p_n changes sign. Then $m \leqslant n$, since p_n has at most n zeros.
2. Assume $m \leqslant n-1$. Put $q(x) := \prod_{i=1}^{m}(x - x_i) \in \prod_m \subset \prod_{n-1}$ ($q \equiv 1$ if $m = 0$).
3. The product has one sign. Each factor $x - x_i$ changes sign at x_i, exactly where p_n does. So $p_n q$ does not change sign on (a, b), and $p_n q \not\equiv 0$. Being continuous,

$$\langle p_n, q \rangle = \int_a^b p_n(x) q(x)\, dx \neq 0.$$

4. Contradiction. Orthogonality gives $\langle p_n, q \rangle = 0$. Hence $m = n$: p_n has n distinct zeros in (a, b), and these are all its zeros, since $\deg p_n = n$. ■

Gauss Nodes for $n = 1, \ldots, 5$

Nodes on $[-1, 1]$; dot area proportional to the weight w_i (weights are derived on the next frames).



- Nodes cluster toward the end points; the weights are largest in the middle.
- $n = 2$: nodes $\pm\frac{1}{\sqrt{3}}$, weights $1, 1$.
- $n = 3$: nodes $0, \pm\sqrt{\frac{3}{5}}$, weights $\frac{8}{9}, \frac{5}{9}, \frac{5}{9}$ (derived below).

Weights: Exactness Is Tested on a Basis

With the nodes fixed, the weights are chosen so that (1) is exact for all $f \in \prod_{n-1}$. Write $Q(f) := \sum_{i=1}^{n} w_i f(x_i)$ and $I(f) := \int_a^b f(x)\,dx$.

Lemma. Let V be a vector space of integrable functions with basis $\varphi_1, \ldots, \varphi_m$. Then

$$Q(f) = I(f) \text{ for all } f \in V \quad \Longleftrightarrow \quad Q(\varphi_j) = I(\varphi_j), \;\; j = 1, \ldots, m.$$

Proof. ($\Rightarrow$) Each φ_j lies in V. ($\Leftarrow$) Every $f \in V$ is $f = \sum_j c_j \varphi_j$. Since Q and I are linear,

$$Q(f) = \sum_{j=1}^{m} c_j Q(\varphi_j) = \sum_{j=1}^{m} c_j I(\varphi_j) = I(f).$$

■

Remark. For $V = \prod_{n-1}$ ($m = n$) this gives n linear equations for $w_1, \ldots, w_n$, one per basis function. Next: two choices of basis.

Weights, Basis 1: The Monomials

Take $\varphi_{j+1}(x) = x^j$, $j = 0, \ldots, n-1$. The conditions $Q(x^j) = I(x^j)$ read $\sum_{i=1}^n w_i x_i^j = \frac{1}{j+1}(b^{j+1} - a^{j+1})$:

$$\begin{bmatrix} 1 & 1 & \cdots & 1 \\ x_1 & x_2 & \cdots & x_n \\ x_1^2 & x_2^2 & \cdots & x_n^2 \\ \vdots & \vdots & \ddots & \vdots \\ x_1^{n-1} & x_2^{n-1} & \cdots & x_n^{n-1} \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \end{bmatrix} = \begin{bmatrix} b-a \\ \frac{1}{2}(b^2 - a^2) \\ \vdots \\ \frac{1}{n}(b^n - a^n) \end{bmatrix}$$

Proposition. If $x_1, \ldots, x_n$ are distinct, the matrix is nonsingular, so the weights exist and are unique.

Proof. It is a Vandermonde matrix, with determinant $\prod_{1 \leqslant i < j \leqslant n}(x_j - x_i) \neq 0$. ■

Weights, Basis 2: The Lagrange Basis

Definition (*Lagrange basis polynomials*). For distinct nodes $x_1, \ldots, x_n$,

$$\ell_i(x) := \prod_{\substack{j=1 \\ j \neq i}}^{n} \frac{x - x_j}{x_i - x_j} \in \prod_{n-1}, \qquad \ell_i(x_k) = \begin{cases} 1 & k = i, \\ 0 & k \neq i. \end{cases}$$

They form a basis of $\prod_{n-1}$: $n = \dim \prod_{n-1}$ functions, and $\sum_i c_i \ell_i = 0$ gives $c_k = 0$ at x_k.

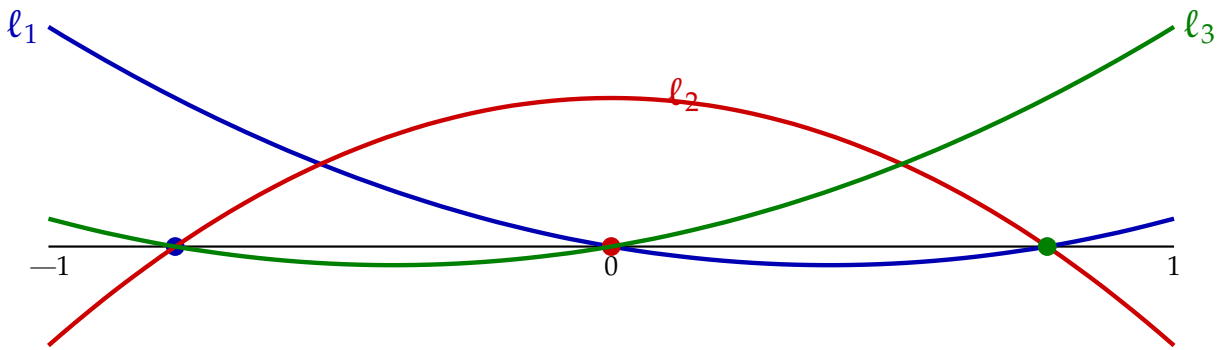
Proposition. Exactness on $\prod_{n-1}$ holds if and only if

$$w_i = \int_a^b \ell_i(x)\, dx, \qquad i = 1, \ldots, n.$$

Proof. By the lemma, exactness means $Q(\ell_k) = I(\ell_k)$ for all k. Since $\ell_k(x_i) = \delta_{ik}$, $Q(\ell_k) = \sum_i w_i \ell_k(x_i) = w_k$, so the condition is $w_k = \int_a^b \ell_k\, dx$. ∎

Weights as Areas: $n = 3$ on $[-1, 1]$

Nodes $-\sqrt{3/5},\ 0,\ \sqrt{3/5}$ (zeros of $p_3 = x^3 - \frac{3}{5}x$).



$$\int \ell_1 = \tfrac{5}{9}, \quad \int \ell_2 = \tfrac{8}{9}, \quad \int \ell_3 = \tfrac{5}{9}$$

w_i is the signed area under ℓ_i. E.g. $\ell_2(x) = \dfrac{x^2 - 3/5}{-3/5} = 1 - \frac{5}{3}x^2$, so

$$w_2 = \int_{-1}^{1} \left(1 - \tfrac{5}{3}x^2\right) dx = 2 - \tfrac{5}{3} \cdot \tfrac{2}{3} = \tfrac{8}{9}.$$

By symmetry $w_1 = w_3$; $w_1 + w_2 + w_3 = 2$ ($f \equiv 1$) gives $w_1 = w_3 = \frac{5}{9}$.

Example: The Two-Point Rule ($n = 2$)

Nodes $x_{1,2} = \mp\frac{1}{\sqrt{3}}$ (zeros of $p_2 = x^2 - \frac{1}{3}$) on $[-1, 1]$. Weights: ℓ_1, ℓ_2 are mirror images, so $w_1 = w_2$, and $w_1 + w_2 = 2$ (from $f \equiv 1$) gives $w_1 = w_2 = 1$. The rule is $Q(f) = f\left(-\frac{1}{\sqrt{3}}\right) + f\left(\frac{1}{\sqrt{3}}\right)$.

f	$\int_{-1}^{1} f\,dx$	$Q(f)$	exact?
1	2	$1 + 1 = 2$	yes
x	0	$-\frac{1}{\sqrt{3}} + \frac{1}{\sqrt{3}} = 0$	yes
x^2	$\frac{2}{3}$	$\frac{1}{3} + \frac{1}{3} = \frac{2}{3}$	yes
x^3	0	$-\frac{1}{3\sqrt{3}} + \frac{1}{3\sqrt{3}} = 0$	yes
x^4	$\frac{2}{5}$	$\frac{1}{9} + \frac{1}{9} = \frac{2}{9}$	no

Degree of exactness $3 = 2n - 1$, with only 2 function values. (Trapezoid, also 2 points: degree 1.)

The Degrees-of-Freedom Heuristic, Audited

Counting: n nodes $+$ n weights $= 2n$ parameters, and dim $\prod_{2n-1} = 2n$. This *suggests* degree $2n-1$, but guarantees nothing; the equations are nonlinear in the nodes. For instance Simpson ($n = 3$, fixed equally spaced nodes) has degree $3 < 5$.
What Gauss achieved:

- Nodes $=$ zeros of p_n *decouple* the nonlinearity: with the nodes fixed, the weights solve a *linear* problem.
- Conclusion (preview; precise form and proof in the next lecture): this rule is exact on $\prod_{2n-1}$.

Proposition (*Degree $2n-1$ cannot be beaten*). No rule of the form (1) with n nodes is exact on $\prod_{2n}$.

Proof. Let $q(x) := \prod_{i=1}^{n}(x - x_i)^2 \in \prod_{2n}$. Then $q \geqslant 0$, $q \not\equiv 0$, so $\int_a^b q\,dx > 0$, while $\sum_i w_i q(x_i) = 0$. ∎

Efficiency, Positive Weights, Computation

Efficiency at $n = 5$. Gauss: degree 9. Newton–Cotes with 5 equally spaced points: degree 5 (interpolation gives 4, symmetry one more, as for Simpson in Lecture 22).

Positive weights. Taking $f \equiv 1$ in (1) gives $\sum_i w_i = b - a$.

Remark (*Conclusion (preview; proof in the next lecture)*). All Gaussian weights are positive.

Then $\sum_i |w_i| = b - a$ for every n: no cancellation, so the rule is stable. Newton–Cotes rules lose positivity of the weights for large n, which is why they are not used at high order.

Computation. In practice, nodes and weights come from an eigenvalue problem for a tridiagonal "Jacobi matrix" (Golub–Welsch), a preview of Lectures 25–29.

Looking Ahead

- **Next lecture:** Gaussian Integration